

Course Code : ECT 359-4

WYXZ/MS – 23 / 1606

Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination

INTRODUCTION TO IOT

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as indicated.
- (2) Due credit will be given to neatness.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1.
 - (a) For a Smart Manufacturing Use Case, recommend an architecture with details and justification. 7(CO1,5)
 - (b) As an IoT Application Developer, select an Access Technology for Smart Waste Management applications. Justify. 3(CO1,5)
2.
 - (a) Compare Raspberry pi with camera setup and ESP32-CAM for a Industrial Robot application. 4(CO1)
 - (b) Describe the steps to expand image size of 16 GB and utilize the available 64 GB on a SD card using RPi. 4(CO2)
3.
 - (a) Write a bash script to execute a script "test1.sh" in a loop with a two input argument stored in arrays of length 4 and 8, respectively. 6(CO2)
 - (b) Write a bash script to clone a, compile, build and install source code of tensorflow located at a git repository "tensorflow/tensorflow" using cmake. 6(CO1,3,4)
4.
 - (a) Write a python script to perform moving average of three elements of an array $X = [1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9]$. 5(CO1,3,4)

WYXZ/MS-23 / 1606

Contd.

- (b) Write a python script to read an analog pressure sensor value deployed on domestic bio gas plant in a remote village using pi to control a valve locally driven by servo, at 40-60 PSI valve should be ON otherwise OFF. 5(CO1,3,4)
5. (a) Write a python script to interface ADC with pi and acquire an analog signal from soil moisture sensor and turn on the water pump. 5(CO1,3,4)
- (b) Compare I2C and SPI protocol with examples. 5(CO4)
6. (a) Identify the challenges of implementing an IoT. Use case for Automated retail shops in India. How would you overcome those ? Justify. 7(CO1,5)
- (b) Explain the XMPP protocol with example. 3(CO1)

